

Lecture 13: Newton's Method, Levenberg-Marquardt, and Quasi-Newton Methods

Instructor Proof Notes

Color Legend

Green: Step numbering and substeps **Orange:** Key concepts **Purple:** Main results **Blue:** Teaching notes

Proof Roadmap

Theorem	Time	Board-Friendly	Priority
Theorem 1: Newton Direction is a Descent Direction	5–8 min	Yes	Full
Theorem 2: Quadratic Convergence of Newton's Method	10–12 min	Moderate	Full
Theorem 3: Newton Solves Quadratics in One Step	3–5 min	Yes	Full
Theorem 4: BFGS Maintains Positive Definiteness	8–10 min	Sketch	Key Result

1 Theorem 1: Newton Direction is a Descent Direction

Statement: Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice continuously differentiable. Suppose $\nabla^2 f(x_k)$ (the Hessian at x_k) is **positive definite**. Then the **Newton direction**

$$d_k = -\nabla^2 f(x_k)^{-1} \nabla f(x_k)$$

is a **descent direction**, meaning $\nabla f(x_k)^T d_k < 0$.

Proof:

Step 1: Define the one-dimensional function along the Newton direction:

$$\theta_k(\lambda) = f(x_k + \lambda d_k).$$

Step 2: Compute the derivative with respect to λ :

$$\theta'_k(\lambda) = \nabla f(x_k + \lambda d_k)^T d_k.$$

Step 3: Evaluate the derivative at $\lambda = 0$:

$$\theta'_k(0) = \nabla f(x_k)^T d_k.$$

Step 4: Substitute the definition of d_k :

$$\theta'_k(0) = \nabla f(x_k)^T (-\nabla^2 f(x_k)^{-1} \nabla f(x_k)) = -\nabla f(x_k)^T \nabla^2 f(x_k)^{-1} \nabla f(x_k).$$

Step 5: Let $H_k = \nabla^2 f(x_k)$ and $g_k = \nabla f(x_k)$. Then,

$$\theta'_k(0) = -g_k^T H_k^{-1} g_k.$$

Step 6: Since H_k is positive definite, H_k^{-1} is also positive definite. Therefore, the quadratic form $g_k^T H_k^{-1} g_k > 0$ (unless $g_k = 0$, i.e., at a critical point).

Step 7: Thus, $\theta'_k(0) = -g_k^T H_k^{-1} g_k < 0$, which means the directional derivative along d_k is negative.

The Newton direction d_k is a descent direction.

Teaching Notes: Theorem 1

Board Strategy: This is a clean, short proof. Write out the definition of $\theta_k(\lambda)$ clearly and show how the chain rule gives the directional derivative.

Key Insight: The Newton direction leverages second-order information (the Hessian). When the Hessian is positive definite, it acts like a “curved metric,” ensuring that the Newton direction points downhill.

Why This Matters: This justifies why we can use the Newton direction in a line search: we are guaranteed to find a step size $\lambda > 0$ that reduces the objective, assuming the Hessian is positive definite locally.

Timing: 5–8 minutes. This proof can be rushed if time is tight; the conclusion is more important than the algebra.

Common Student Confusion

Q: “What if the Hessian is not positive definite? Can Newton’s method still work?”

A: If the Hessian is indefinite or negative definite, the Newton direction may point uphill or be ill-defined (singular Hessian). This is why practical implementations use *modifications* like Levenberg-Marquardt (adding a damping term) or quasi-Newton methods (approximating with a positive definite matrix). Newton’s method in pure form assumes a positive definite Hessian.

2 Theorem 2: Quadratic Convergence of Newton’s Method

Statement: Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is twice continuously differentiable in a neighborhood of a point $x^* \in \mathbb{R}^n$. Assume:

- (i) $\nabla f(x^*) = 0$ (i.e., x^* is a critical point),
- (ii) $\nabla^2 f(x^*)$ is positive definite,
- (iii) The Hessian $\nabla^2 f$ is **Lipschitz continuous** in a neighborhood of x^* , with constant $L > 0$.

Then, starting from a point x_0 sufficiently close to x^* , Newton's method with exact Hessian inversion converges **quadratically**: there exist constants $C > 0$ and $\delta > 0$ such that if $\|x_0 - x^*\| < \delta$, then

$$\|x_{k+1} - x^*\| \leq C\|x_k - x^*\|^2.$$

Proof Sketch:

Step 1: Define the error $e_k = x_k - x^*$. The Newton iterate is

$$x_{k+1} = x_k - \nabla^2 f(x_k)^{-1} \nabla f(x_k).$$

Step 2: By Taylor expansion around x^* ,

$$\nabla f(x_k) = \nabla f(x^*) + \nabla^2 f(x^*)e_k + O(\|e_k\|^2).$$

Since $\nabla f(x^*) = 0$ and assuming second-order error term $O(\|e_k\|^2)$ captures Lipschitz variation:

$$\nabla f(x_k) = \nabla^2 f(x^*)e_k + O(\|e_k\|^2).$$

Step 3: Rearrange the Newton step:

$$e_{k+1} = e_k - \nabla^2 f(x_k)^{-1} (\nabla^2 f(x^*)e_k + O(\|e_k\|^2)).$$

Step 4: Denote $\Delta H_k = \nabla^2 f(x_k) - \nabla^2 f(x^*)$. By Lipschitz continuity, $\|\Delta H_k\| \leq L\|e_k\|$.

Step 5: Write

$$\nabla^2 f(x_k) = \nabla^2 f(x^*) + \Delta H_k.$$

The inverse satisfies (using perturbation analysis):

$$\nabla^2 f(x_k)^{-1} = (\nabla^2 f(x^*) + \Delta H_k)^{-1} \approx \nabla^2 f(x^*)^{-1} - \nabla^2 f(x^*)^{-1} \Delta H_k \nabla^2 f(x^*)^{-1} + O(\|e_k\|^2).$$

Step 6: Substitute back into Step 3 and simplify:

$$e_{k+1} = e_k - [\nabla^2 f(x^*)^{-1} - \nabla^2 f(x^*)^{-1} \Delta H_k \nabla^2 f(x^*)^{-1} + O(\|e_k\|^2)] [\nabla^2 f(x^*)e_k + O(\|e_k\|^2)].$$

Step 7: The leading term is:

$$\begin{aligned} e_{k+1} &\approx e_k - \nabla^2 f(x^*)^{-1} \nabla^2 f(x^*)e_k + \nabla^2 f(x^*)^{-1} \Delta H_k \nabla^2 f(x^*)^{-1} \nabla^2 f(x^*)e_k + O(\|e_k\|^2) \\ &= e_k - e_k + \nabla^2 f(x^*)^{-1} \Delta H_k e_k + O(\|e_k\|^2). \end{aligned}$$

Step 8: Since $\|\Delta H_k\| \leq L\|e_k\|$, we have

$$\|e_{k+1}\| \lesssim \|\nabla^2 f(x^*)^{-1}\| \cdot L\|e_k\| \cdot \|e_k\| + O(\|e_k\|^2) \sim C\|e_k\|^2.$$

For sufficiently small ϵ_0 , Newton's method converges quadratically with rate constant $C \approx \frac{L}{2} \|\nabla^2 f(x^*)\|$

Teaching Notes: Theorem 2

Board Strategy: This proof is technical; present the overall structure clearly. Emphasize that the Lipschitz Hessian assumption is what drives the quadratic error bound.

Key Insight: Quadratic convergence is a gold standard: once you are close enough to the optimum, the error squares at each iteration. With quadratic convergence, 50 iterations can be enough for very high accuracy (doubling correct digits each iteration).

Why Lipschitz Hessian? The Lipschitz condition on $\nabla^2 f$ ensures that the Hessian does not vary “too wildly” near x^* . This is the technical condition that prevents the Hessian perturbation from dominating the Newton step.

Local vs. Global: This theorem is a *local* convergence result. It guarantees fast convergence only if you start close to x^* . For global convergence, use line search.

Timing: 10–12 minutes. The sketch version is clearer than the full rigorous proof.

Common Student Confusion

Q: “If Newton's method converges quadratically, why not always use it?”

A: Three reasons: (1) It requires the Hessian (expensive to compute for large n). (2) The Hessian may not be positive definite away from the optimum. (3) The quadratic convergence is only *local*—you must start close to the optimum. For global convergence, you need to combine Newton with line search or trust regions.

Q: “What is ‘sufficiently close’ in the theorem?”

A: The radius of convergence depends on the Lipschitz constant L and the condition number of $\nabla^2 f(x^*)$. For well-conditioned problems, it can be reasonably large; for ill-conditioned problems, it can be tiny. In practice, a hybrid approach (gradient descent to get close, then Newton) is often used.

3 Theorem 3: Newton's Method Solves Quadratics in One Step

Statement: Consider the quadratic function $f(x) = \frac{1}{2}x^T Qx + c^T x + d$ where $Q \in \mathbb{R}^{n \times n}$ is **positive definite** and $c \in \mathbb{R}^n$, $d \in \mathbb{R}$. Starting from any point $x_0 \in \mathbb{R}^n$, one iteration of Newton's method yields the **global minimizer** x^* .

Proof:

Step 1: Compute the gradient:

$$\nabla f(x) = Qx + c.$$

Step 2: Compute the Hessian:

$$\nabla^2 f(x) = Q.$$

Note that the Hessian is constant (independent of x).

Step 3: The Newton step at x_0 is:

$$x_1 = x_0 - Q^{-1}(Qx_0 + c) = x_0 - x_0 - Q^{-1}c = -Q^{-1}c.$$

Step 4: The global minimizer x^* satisfies $\nabla f(x^*) = 0$:

$$Qx^* + c = 0 \quad \Rightarrow \quad x^* = -Q^{-1}c.$$

Step 5: Therefore, $x_1 = x^*$.

Newton's method converges to the global minimizer of a quadratic in exactly one step, regardless of the starting point.

Teaching Notes: Theorem 3

Board Strategy: This is a quick, satisfying proof. Write the quadratic, compute gradient and Hessian (emphasizing that Q is constant), and show that the Newton step lands exactly at the optimum.

Key Insight: For quadratic functions, the Newton method is not just fast—it is *exact*. This is because Newton's method solves the quadratic Taylor model locally, and for quadratics, the local model *is* the function itself.

Why It Matters: This theorem explains why Newton's method has such fast convergence near the optimum of general functions: locally, any smooth function looks like a quadratic, and the Newton model is “very accurate” there.

Timing: 3–5 minutes. This is a satisfying warm-up after the more complex Theorem 2.

Common Student Confusion

Q: “If Newton's method solves quadratics in one step, why does it take longer for non-quadratic functions?”

A: Because the Newton model (which uses the Hessian as if it were constant like a quadratic) is only an approximation for general functions. Far from the optimum, the Hessian changes significantly, so the quadratic model is inaccurate. As you approach the optimum, the model becomes accurate, and convergence accelerates (quadratically).

Q: “What if we start from the same x_0 but use a different algorithm (e.g., gradient descent) on the quadratic?”

A: Gradient descent converges but takes multiple iterations, proportional to the condition number of Q . The convergence rate is geometric, not finite. Newton's finite-convergence property is very special to Newton and is only possible because it uses exact second-order information.

4 Theorem 4: BFGS Maintains Positive Definiteness

Statement: The **Broyden-Fletcher-Goldfarb-Shanno (BFGS)** quasi-Newton update is defined by

$$B_{k+1} = B_k - \frac{B_k s_k s_k^T B_k}{s_k^T B_k s_k} + \frac{y_k y_k^T}{s_k^T y_k},$$

where $s_k = x_{k+1} - x_k$ is the step and $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$ is the gradient change. If B_k is **positive definite** and the **secant condition** $s_k^T y_k > 0$ holds, then B_{k+1} is also **positive definite**.

Proof Sketch:

Step 1: Rewrite the BFGS update in a more compact form. Define

$$P = I - \frac{s_k y_k^T}{y_k^T s_k}, \quad Q = \frac{y_k y_k^T}{y_k^T s_k}.$$

Then,

$$B_{k+1} = P^T B_k P + Q.$$

Step 2: Consider any nonzero vector $v \in \mathbb{R}^n$. We compute $v^T B_{k+1} v$:

$$v^T B_{k+1} v = v^T P^T B_k P v + v^T Q v.$$

Step 3: The second term: Since $Q = \frac{y_k y_k^T}{y_k^T s_k}$ and $s_k^T y_k > 0$, the matrix Q is a positive semi-definite rank-one update. Thus,

$$v^T Q v = \frac{(y_k^T v)^2}{y_k^T s_k} \geq 0.$$

Step 4: The first term: Since B_k is positive definite, for any vector w , we have $w^T B_k w > 0$ (assuming $w \neq 0$). Now, $w = P v$:

$$v^T P^T B_k P v = (P v)^T B_k (P v) \geq 0.$$

Step 5: To show strict positivity, suppose $v^T B_{k+1} v = 0$. Then both $v^T P^T B_k P v = 0$ and $v^T Q v = 0$ must hold.

Step 6: From $v^T Q v = 0$, we get $(y_k^T v)^2 = 0$, so $y_k^T v = 0$.

Step 7: From $v^T P^T B_k P v = 0$ and positive definiteness of B_k , we must have $P v = 0$, i.e.,

$$\left(I - \frac{s_k y_k^T}{y_k^T s_k} \right) v = 0 \quad \Rightarrow \quad v = \frac{s_k y_k^T v}{y_k^T s_k}.$$

Step 8: But we also have $y_k^T v = 0$ from Step 6. Substituting:

$$v = \frac{s_k \cdot 0}{y_k^T s_k} = 0.$$

Step 9: Contradiction: we assumed $v \neq 0$. Therefore, $v^T B_{k+1} v > 0$ for all $v \neq 0$.

The BFGS update B_{k+1} is positive definite whenever B_k is positive definite and $s_k^T y_k > 0$.

Teaching Notes: Theorem 4

Board Strategy: This proof involves matrix identities; write the update rule clearly and walk through the reasoning step-by-step. The key steps are identifying the rank-one structure and using positive definiteness of B_k carefully.

Key Insight: The BFGS update is carefully designed so that positive definiteness is *hereditary*—if you start with a positive definite approximation B_0 (e.g., $B_0 = I$), all subsequent updates remain positive definite. This is crucial because it ensures descent directions.

The Secant Condition: The condition $s_k^T y_k > 0$ is ensured by using a line search that satisfies the strong Wolfe conditions. Without it, BFGS can break down. This is why backtracking line search is paired with quasi-Newton methods.

Why Two Terms? The BFGS formula subtracts one rank-one term $(-\frac{B_k s_k s_k^T B_k}{s_k^T B_k s_k})$, from the bad Broyden family) and adds another $(\frac{y_k y_k^T}{y_k^T s_k})$, which enforces the secant equation). The combination is carefully balanced to maintain positive definiteness.

Timing: 8–10 minutes for a sketch. Full details can be assigned as reading.

Common Student Confusion

Q: “Why do we need the secant condition $s_k^T y_k > 0$? Can’t we just use any step and gradient pair?”

A: The secant condition is *induced* by a suitable line search. When you use the strong Wolfe conditions, they automatically ensure $s_k^T y_k > 0$. If you relax the line search or use poor step selection, this can fail, and BFGS loses the positive definiteness guarantee. In practice, always pair BFGS with a line search.

Q: “Why does BFGS use two rank-one terms instead of just one?”

A: The one-rank-one update (Broyden’s method) can lose positive definiteness. BFGS is designed with two rank-one terms such that the “correction” (the second term) dominates and maintains positive definiteness while also satisfying an important quasi-Newton property (the secant equation). It is an elegant balancing act.

Q: “How is B_k initialized? Does it affect convergence?”

A: Typically, $B_0 = I$ (identity). This is positive definite and ensures that the first step is a scaled gradient descent step. Poor initialization (e.g., B_0 singular or nearly singular) slows convergence, but BFGS still works as long as B_0 is positive definite and the line search maintains $s_k^T y_k > 0$.

Summary and Connections

The Big Picture:

1. **Theorem 1** establishes that Newton's direction is downhill (descent) when the Hessian is positive definite.
2. **Theorem 2** shows that Newton's method converges *quadratically* near the optimum, provided the Hessian is Lipschitz continuous. This is the gold standard for unconstrained optimization.
3. **Theorem 3** reveals a special case: for quadratic functions, Newton solves the problem in *exactly one step*. This explains why Newton works so fast near the optimum (where the function is locally quadratic).
4. **Theorem 4** proves that BFGS, a practical quasi-Newton method, preserves positive definiteness of the Hessian approximation. This ensures we always have a descent direction, even without computing true Hessians.

Why This Matters: Newton's method is the “gold standard” for smooth optimization: quadratic convergence is unbeatable. However, computing Hessians is expensive. Quasi-Newton methods like BFGS provide a cheap approximation that maintains key properties (descent directions, positive definiteness) and achieves superlinear convergence without full Hessian information.

Practical Implementation Notes:

- **Pure Newton:** Requires computing $\nabla^2 f(x_k)$ and solving $\nabla^2 f(x_k)d_k = -\nabla f(x_k)$ at each iteration. For $n > 1000$, this can be prohibitive.
- **Newton with line search:** Pair with a backtracking line search or strong Wolfe conditions to ensure global convergence. Without line search, Newton can diverge if the Hessian is not positive definite away from the optimum.
- **Levenberg-Marquardt (LM):** Modifies Newton by adding a damping term: $(H_k + \mu I)d_k = -\nabla f(x_k)$. Automatically blends Newton (small μ) with gradient descent (large μ). Popular for nonlinear least squares problems.
- **BFGS (Quasi-Newton):** Avoids computing Hessians. Instead, builds a positive definite approximation B_k using only gradient information. Achieves superlinear convergence. Default choice for medium-scale problems ($n \sim 100$ – 10000).
- **Limited-Memory BFGS (L-BFGS):** Stores only the last m updates (e.g., $m = 20$) instead of the full $n \times n$ matrix. Suitable for large-scale problems ($n > 10000$). Same superlinear convergence but with $O(mn)$ storage instead of $O(n^2)$.

Recommended Homework / Discussion

- Verify Theorem 1 for a specific function (e.g., $f(x) = x_1^2 + 4x_2^2$) and confirm that the Newton direction is a descent direction.
- Implement Newton's method with exact Hessian on a small quadratic problem ($n = 2$ or $n = 3$) and verify one-step convergence (Theorem 3).
- Implement BFGS for a standard test function (e.g., Rosenbrock) and compare number of iterations vs. gradient descent and Newton's method.
- Discuss: why does Levenberg-Marquardt work better than pure Newton on noisy least-squares problems?
- Derive the Sherman-Morrison-Woodbury formula for efficient BFGS updates (avoiding explicit matrix inversion).